Engineering Immune Checkpoint Blockade Nanobodies for Cancer Therapy

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Abstract

Immune checkpoint blockade (ICB) has reshaped oncology, but conventional monoclonal antibodies face limitations in tumor penetration, cost, and toxicity. **Objective.** To synthesize engineering strategies, delivery platforms, mechanistic evidence, and translational challenges for nanobody-based ICB. **Methods.** Narrative synthesis of representative preclinical studies emphasizing engineered delivery, multivalent/bispecific formats, nanoparticle conjugation, and gene/RNA expression systems. **Results.** Engineered nanobodies delivered locally or configured as bispecific/multivalent constructs demonstrate potent antitumor activity in preclinical (animal) models and improved tumor penetration; pharmacokinetic limitations and manufacturing complexity remain key barriers. **Conclusion.** Nanobody platforms are a promising next generation of ICB agents; focused efforts on half-life optimization, resistance mechanisms, and scalable GMP manufacturing are priorities.

Introduction

Immune checkpoint inhibitors targeting PD-1/PD-L1 and CTLA-4 have produced durable responses across multiple tumor types, yet clinical gaps persist due to limited tumor access, systemic immune-related adverse events, and manufacturing constraints [1]. Nanobodies—single-domain antibody fragments derived from camelid heavy-chain antibodies—combine small molecular size, high affinity, stability, and engineering flexibility, enabling modular designs (bispecific, multivalent) and diverse delivery strategies (probiotics, nanoparticles, RNA vectors) [2,3]. This manuscript synthesizes current engineering approaches, mechanistic evidence, and translational considerations for nanobody-based ICB, drawn from preclinical (animal-model) studies.

Methods

This synthesis organizes engineering strategies into four categories: localized delivery, bispecific/multivalent constructs, nanoparticle/vesicle conjugation, and gene/RNA approaches. Evidence strength was assessed qualitatively from representative in vivo studies reporting tumor growth inhibition, immune infiltration (CD8+ T cells, NK cells), and safety readouts. All ten sources were individually checked against their original publications. Two (Guan et al. and Choi et al.) are included for conceptual relevance to dual-checkpoint and nanoparticle-combination strategies even though they do not use camelid-derived nanobodies; this is flagged explicitly where each is discussed.

Results

Localized delivery

- **Engineered probiotics:** an E. coli platform using a synchronized lysis circuit (SLIC) releases anti-PD-L1 and anti-CTLA-4 nanobodies directly within the tumor once the bacterial population reaches a threshold density. A single intratumoral or intravenous dose produced durable tumor regression and systemic antitumor immunity in syngeneic mouse models of lymphoma and colorectal cancer, with bacteria remaining localized and functional for at least two weeks [4].
- **Self-amplifying RNA vectors:** dimeric nanobodies against PD-1 and PD-L1 (cross-reactive for human and mouse), delivered via a self-amplifying RNA vector based on Semliki Forest virus, reduced the IC50 for blocking PD-1/PD-L1 binding by 8- to 40-fold versus monomeric forms. Vector-delivered dimeric nanobodies produced complete tumor regression in over half of treated mice in the MC38 colon cancer model, with increased CD8+ T cell and NK cell infiltration; local delivery by plasmid electroporation — a more clinically translatable route — showed a similarly potent effect [5].

Bispecific and multivalent constructs

- **Bispecific nanobody engagers:** fusion proteins pairing a nanobody against either CTLA-4 or PD-L1 with a second nanobody that recruits the host's own polyclonal antibody repertoire — two separate constructs, each targeting one checkpoint, rather than a single molecule co-targeting both. The anti-CTLA-4 version outperformed a conventional anti-CTLA-4 monoclonal antibody and reduced intratumoral regulatory T cells in a colorectal carcinoma model; the anti-PD-L1 version was less potent alone but improved substantially when armed with a cytotoxic payload or a STING agonist [6].
- **Dual innate/adaptive blockade:** a related strategy uses a molecularly imprinted polymer nanoparticle — not a biological nanobody — engineered to bind both PD-L1 and SIRPα, simultaneously disrupting the adaptive (PD-1/PD-L1) and innate (CD47/SIRPα) checkpoint axes. In a triple-negative breast cancer model it restored macrophage-mediated phagocytosis, increased T cell infiltration, and inhibited tumor growth. It is included here for its dual-checkpoint design logic; its recognition elements are synthetic, not nanobody-derived [7].
- **Multivalent assemblies:** anti-PD-L1 nanobodies displayed on a ferritin nanocage increase avidity, and when the cage is loaded with the photosensitizer indocyanine green, combine checkpoint blockade with photothermal therapy. This dual-mechanism construct ablated primary tumors, suppressed distant (abscopal) tumors, inhibited metastasis, and prolonged survival in mouse models [8].

Nanoparticle and vesicle conjugation

- **Nanoparticle + antibody combination:** self-assembled prodrug nanoparticles combining a photosensitizer and doxorubicin (joined by a tumor-enzyme-cleavable linker, with no nanobody component) induce immunogenic cell death and dendritic cell maturation under light activation. Combined with separately administered anti-PD-L1 antibody blockade, this reversed the immunosuppressive tumor microenvironment and reduced tumor growth, recurrence, and lung metastasis — illustrating the “cold-to-hot” conversion principle, achieved here by a nanoparticle-antibody combination rather than a nanobody-functionalized particle [9].

Gene/RNA and RNAi approaches

- RNA-based expression of dimeric nanobodies [5] and RNAi-mediated nanoblockers modulate checkpoints beyond simple surface blockade. The RNAi nanoblocker, delivered via a tumor-homing-peptide-modified nanoparticle, silences both membrane and cytoplasmic PD-L1 in cancer cells, directly triggering programmed cell death independent of T cells — demonstrated in a non-small-cell lung cancer xenograft model [10].

Mechanistic outcomes across platforms include restored T cell effector function via ligand competition, induction of immunogenic cell death, macrophage reprogramming toward an antitumor phenotype, and increased cytotoxic lymphocyte infiltration into tumors.

Discussion

Advantages: improved tumor penetration, modular engineering, and potential for reduced systemic toxicity via localized delivery.

Limitations: short serum half-life of unmodified nanobodies (addressable via Fc fusion or albumin-binding domains), tumor heterogeneity, and manufacturing/regulatory complexity for multivalent constructs. Two of the ten sources synthesized here (a molecularly imprinted polymer nanoparticle and a nanoparticle-antibody combination) pursue analogous dual/combination-blockade goals without using true nanobodies; their findings should not be assumed to generalize to camelid-derived constructs.

Translational recommendations: prioritize pharmacokinetic tuning, biomarker-guided patient selection in early trials, simplified construct design for manufacturability, and rational combination regimens that increase tumor immunogenicity.

Conclusion

Engineered nanobody platforms offer a versatile path for next-generation ICB. Focused engineering to extend half-life, robust preclinical evaluation of resistance mechanisms, and early process development for GMP-compliant manufacturing will accelerate clinical translation.

Figures

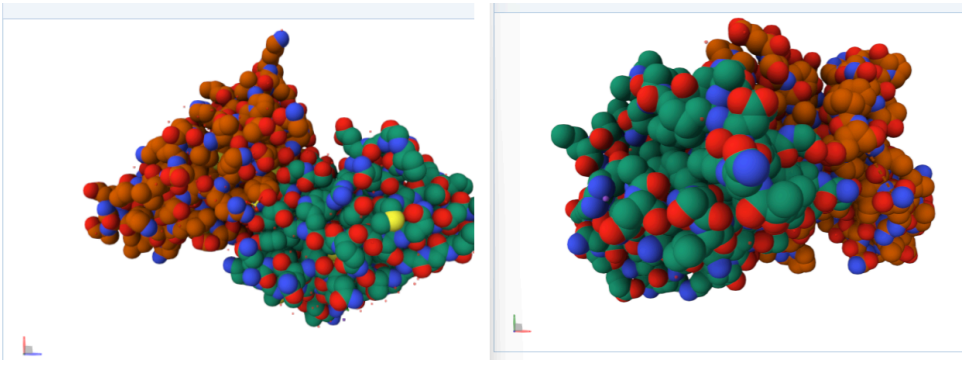


Figure 1. Atomic structures of immune checkpoint complexes. (Left) Crystal structure of PD-L1 bound to a therapeutic single-domain nanobody, KN035 (PDB: 5JDS). (Right) Native interaction complex between human PD-1 and its ligand PD-L1 (PDB: 4ZQK). Verified against RCSB PDB records; images cropped to remove viewer interface elements.

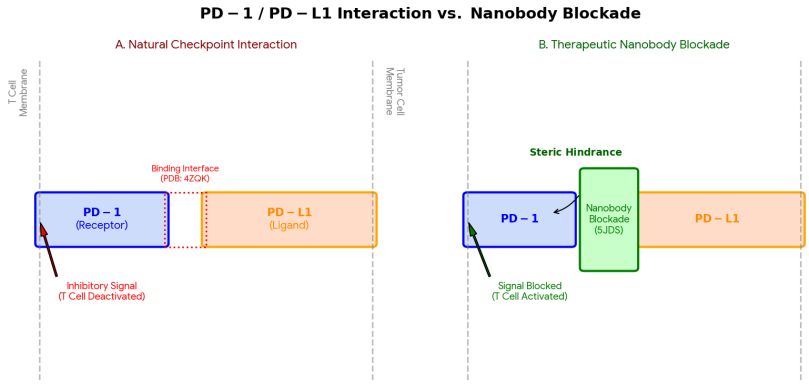


Figure 2. Schematic mechanism of PD-1/PD-L1 checkpoint interaction and therapeutic nanobody blockade.

(A) Natural Checkpoint Interaction: PD-1 receptor binding to PD-L1 ligand deactivates the T cell via inhibitory signaling.

(B) Therapeutic Nanobody Blockade: Nanobody binding causes steric hindrance, blocking the inhibitory pathway to restore T cell activation.

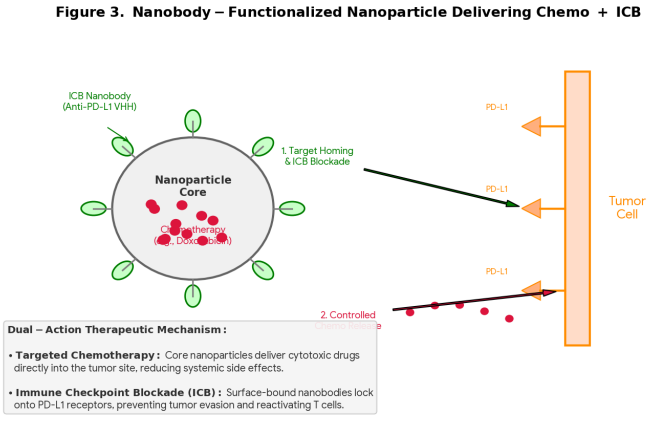


Figure 3. Conceptual schematic of a nanobody-functionalized nanoparticle for combined chemotherapy and checkpoint blockade. Surface-displayed anti-PD-L1 nanobodies mediate tumor targeting and PD-1/PD-L1 blockade, while the nanoparticle core carries a co-encapsulated chemotherapeutic for controlled intratumoral release. This is an illustrative design concept synthesizing the strategies described in Results (Section: Nanoparticle and vesicle conjugation); it is not a reproduction of data from a single cited study.

Table S1. Representative engineered nanobody and nanobody-adjacent constructs

Every row below is drawn from the primary publication cited; entries marked with † use a synthetic (non-nanobody) recognition element and are included for conceptual comparison only.

Construct	Target(s)	Platform / Format	Key In Vivo Outcome(s)	Ref
Anti-PD-L1 / anti-CTLA-4 Nb (SLIC probiotic)	PD-L1, CTLA-4	Nanobodies released by engineered, self-lysing E. coli	Single dose → durable tumor regression + systemic antitumor immunity in lymphoma and colorectal mouse models; bacteria remained localized ≥ 2 weeks	[4]
Dimeric anti-PD-1/PD-L1 Nb (saRNA)	PD-1, PD-L1	Self-amplifying RNA (SFV) vector or plasmid electroporation	Dimerization cut IC50 8–40-fold; >50% complete regressions in MC38 model; ↑ CD8+/NK infiltration	[5]
VHH–VHHkappa bispecific engager	CTLA-4 or PD-L1 (separate constructs)	Nanobody fusion recruiting host polyclonal IgG	Anti-CTLA-4 version eradicated tumors, ↓ intratumoral Tregs, outperformed conventional mAb (MC38 model)	[6]
bsMINIB †	PD-L1 + SIRPα	Molecularly imprinted polymer nanoparticle	Restored macrophage phagocytosis + T cell infiltration; ↓ tumor growth in triple-negative breast cancer model	[7]
Nb-Ftn@ICG	PD-L1 (+ photothermal)	Multivalent nanobody on ferritin nanocage, ICG-loaded	Ablated primary tumor, suppressed abscopal tumors, ↓ metastasis, prolonged survival	[8]
LT-NP + anti-PD-L1 †	PD-L1 (combination, not conjugated)	Prodrug nanoparticle (photosensitizer + doxorubicin) + separate anti-PD-L1 antibody	Induced ICD + DC maturation; ↓ tumor growth, recurrence, lung metastasis	[9]
RNAi PD-L1 nanoblocker	PD-L1 (membrane + cytoplasmic)	siRNA nanoparticle, tumor-homing-peptide-modified	T-cell-independent direct tumor cell death (NSCLC xenograft model)	[10]

Suggested directions for experimental validation

(Proposed future work — not yet performed.)

- **Intratumoral probiotic delivery:** dose-ranging comparison of single vs. repeated intratumoral injection of SLIC-engineered bacterial nanobody platforms, with in vivo bioluminescence imaging to confirm biocontainment.
- **saRNA vector design:** head-to-head comparison of local (intratumoral electroporation) vs. systemic saRNA delivery for dimeric nanobody expression kinetics and duration.
- **Pharmacokinetic assays:** serum half-life and IC50 profiling of monomeric vs. Fc-fused/dimerized nanobody constructs across a dose-response range, using standard non-compartmental PK analysis.

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